

The Trinity-Infinity Framework, Series II: Extended Formalization and Further Illustrative Applications

Revised Edition

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Revision note. This edition revises the original Series II preprint (Nemoto, 2026). Principal changes: (1) Chapter 2 no longer reproduces Series I verbatim; it cites the revised Series I instead, following normal academic practice. (2) The results in Chapter 3 are rebuilt on the corrected operator from the revised Series I and are genuinely extended (anisotropic per-coordinate blending) rather than restated with heavier notation; the stability analysis is rewritten for discrete time, correcting the original's use of a continuous-time Lyapunov derivative. (3) The game-theory and engineering applications (Chapters 4, 6) are extended with a concrete, fully worked example each, computed and verified in code. (4) Numerical validation (Chapter 7) reports only results actually computed for this edition. (5) The original Chapter 7, "Pathways to Societal Implementation and Scalable Governance," claimed mathematically guaranteed applications to AI alignment, climate policy, and governance without derivation; it is replaced by Chapter 8, which discusses the same ideas explicitly as unproven, speculative directions. Prepared with drafting and formalization assistance from Claude (Anthropic); see Acknowledgments.

Abstract

This paper extends the triadic recursion operator introduced in the revised first series of the Trinity-Infinity framework (Nemoto, 2026). We generalize the operator to allow independent blending strength per coordinate and prove this generalized operator retains a unique fixed point, with contraction rate equal to the largest coordinate-wise blend parameter (Theorem 1). We give a direct discrete-time stability argument as an immediate corollary, correcting an earlier continuous-time formulation. We extend the game-theoretic illustration by characterizing, via the folk theorem, the feasible, individually-rational set of payoffs sustainable as subgame-perfect equilibria as the discount factor approaches one, and extend the engineering illustration with a fully worked three-node spring-network example verified against its closed-form Neumann series. All numerical claims are computed and included as a companion script. We close by discussing, explicitly as unproven speculation rather than established results, directions in which this kind of triadic recursive structure might be relevant to governance and AI alignment.

1. Introduction

This paper is the second in a series developing the Trinity-Infinity framework, a conceptual lens organized around a single proven convergence result for a triadic recursion operator [1]. Here we genuinely extend that result — to an anisotropic blending rule with a correspondingly sharper contraction bound — and use the extended operator to deepen, without overstating, the same three illustrative

connections as the first series: repeated games, first-order logic, and modular engineering.

1.1 Position in the Literature

The framework draws on established work in infinitely repeated games [2], triadic relations in semiotics [3], fractal geometry and self-similar structures [4], and complex adaptive systems [5, 6]. We do not extend these literatures; where we use their results (the folk theorem in Section 4, self-similar convergence in Section 6), we compute specific, verifiable instances rather than claiming new general theorems.

2. Summary of Series I

Series I [1] defines the state space $S=[0,1]^3$, the cyclic permutation σ , and the recursion $f(T) = \alpha\sigma(T) + (1-\alpha)p$ for a fixed anchor p and blend $\alpha \in (0,1)$, and proves this has a unique fixed point reached at geometric rate α^n from any initial state (Theorem 1 of Series I). It also gives a closed-form critical discount factor ($\delta^*=1/2$) for cooperation in a three-player repeated game, and flags — rather than claims to solve — analogous questions in logic and engineering. We take this operator as given and extend it below; we do not reproduce its proof here, referring the reader to the original.

3. Extended Formalization

3.1 Generalized Integration Operator

The blend parameter α in Series I applies uniformly to all three coordinates. We generalize to an anisotropic blend $a=(a_1, a_2, a_3)$, each $a_i \in (0,1)$, possibly distinct, defining

$$f(T)_i = a_i \sigma(T)_i + (1-a_i)p_i \text{ for } i=1,2,3.$$

Theorem 1. For every $p \in S$ and every $a=(a_1, a_2, a_3)$ with each $a_i \in (0,1)$, f maps S into itself and satisfies $\|f(x)-f(y)\| \leq \max_i(a_i)\|x-y\|$ for all $x, y \in S$, with the bound tight. Consequently f has a unique fixed point $T^*(p, a) \in S$, reached at geometric rate $\max_i(a_i)^n$ from any initial state.

Proof. As in Series I, S is invariant under f since each coordinate is a convex combination of values in $[0,1]$. For the Lipschitz bound: $f(x)_i - f(y)_i = a_i(\sigma(x)_i - \sigma(y)_i)$, so $|f(x)_i - f(y)_i| \leq \max_i(a_i) \cdot |\sigma(x)_i - \sigma(y)_i|$; summing in quadrature and using that σ is an isometry gives $\|f(x)-f(y)\| \leq \max_i(a_i)\|x-y\|$. Since S is complete and $\max_i(a_i) < 1$, Banach's theorem gives existence, uniqueness, and the stated rate; the bound is tight along the coordinate achieving the maximum. $\square$

We verified this computationally: for $p=(1,0,0.5)$ and $a=(0.5,0.7,0.3)$, six independent random initial states all converged — to within floating-point precision, max deviation 1.1×10^{-16} — to the identical closed-form fixed point $T^*=(0.754190, 0.527933, 0.508380)$. An empirical search over 20,000 random point pairs found a supremum contraction ratio of 0.699973 against a predicted bound of $\max(a_i)=0.7$, matching Theorem 1 to four decimal places.

3.2 Discrete-Time Stability

Since f is a genuine contraction (Theorem 1), stability follows immediately without appeal to continuous-time tools. Let $V(x) = \|x - T^*\|^2$. Because $T^* = f(T^*)$,

$$V(x_{n+1}) \leq \max_i (a_i)^2 V(x_n)$$

so V decreases geometrically at every step, for every initial condition, with no additional hypotheses. The original preprint invoked a continuous-time Lyapunov derivative ($dV/dt \leq -\beta V$) and LaSalle's invariance principle for what is a discrete-time system; LaSalle's principle is built for settings without a closed-form contraction bound, which is not our situation here, and continuous derivatives are not defined for a map iterated in discrete steps. The corrected, discrete statement above is what stability actually requires, and is a direct corollary of Theorem 1 rather than a separate result.

4. Further Applications in Game Theory

Series I derived the discount-factor threshold ($\delta^* = 1/2$) for grim-trigger strategies to sustain mutual cooperation (3,3,3) in the eight-outcome game defined there. We now characterize the full set of payoffs achievable in some subgame-perfect equilibrium as $\delta \rightarrow 1$.

Each player's minmax payoff — the worst payoff the other two can hold a player to, anticipating that player's best response — is 2, identically for all three players by the game's symmetry: since defection strictly dominates for the player being minmaxed, that player always best-responds with D; the other two then minimize the resulting payoff by both defecting, giving $\pi_D(0) = 2$. The feasible set is the convex hull of the eight stage payoffs; we computed this hull directly — all eight outcomes are extreme points, with hull volume 4.00 in payoff space. By the folk theorem [Fudenberg & Maskin, 1986], every payoff profile in this hull with all three coordinates strictly above 2 is sustainable as a subgame-perfect equilibrium payoff for δ sufficiently close to 1. The single grim-trigger threshold from Series I is the simplest point in this set — (3,3,3), the unique payoff-sum-maximizing vertex — not the full picture: profiles such as (2,4,2) and their convex combinations with other individually-rational points are also eventually sustainable, generally at different, higher discount-factor thresholds that a complete case-by-case characterization (beyond this paper's scope) would derive individually.

5. Further Applications in Mathematical Logic

As in Series I, we do not extend the Löwenheim-Skolem theorem or supply a new triadic proof-theoretic construction here. The anisotropic generalization of Section 3.1 does suggest a slightly sharper version of the informal analogy noted in Series I: just as the blend parameters a_i may differ per coordinate, a Henkin-style witnessing construction could in principle assign different priority to different axiom schemas during witness generation. We flag this only as a refinement of the same analogy, not as a step toward an actual construction.

6. Further Applications in Mechanical Engineering

We work a concrete instance of the convergence logic used throughout this paper. Consider three masses arranged in a cycle, each connected to both neighbors by a unit-stiffness spring; the resulting stiffness matrix is

$$K = [[2, -1, -1], [-1, 2, -1], [-1, -1, 2]]$$

— a graph Laplacian for the 3-cycle, a literal instance of a “trinity” structure in a mechanical system. K 's eigenvalues are 0, 3, 3. Choosing $\alpha=0.25$ gives spectral radius $\rho(\alpha K)=0.75 < 1$, satisfying the Neumann-series convergence condition from Series I. We computed the partial sum of the series $K+\alpha K^2+\alpha^2 K^3+\dots$ to 200 terms and compared it against the closed form $K(I-\alpha K)^{-1}$; the two agree to within 3×10^{-15} (floating-point precision), confirming

$$K_\infty = [[8, -4, -4], [-4, 8, -4], [-4, -4, 8]]$$

for this system. This is a fully specified, verified instance rather than a citation of the general identity; we make no claim about its relevance to any specific physical robotic module beyond illustrating that the same contraction logic from Section 3 governs this classical engineering computation too.

7. Numerical Validation

All numerical claims in this paper were computed by executing the operators and matrices exactly as defined in the text; the companion script `series2_verification.py`, distributed with this PDF, reproduces every number reported: the generalized operator's fixed point and empirical contraction ratio (Section 3.1), the convex hull and minmax values (Section 4), and the Neumann series comparison (Section 6). No claim in this paper rests on an unreported simulation.

8. Discussion: Speculative Extensions and Future Directions

The original Series II preprint devoted a full chapter to “Pathways to Societal Implementation and Scalable Governance,” claiming a mathematically guaranteed translation of the framework to AI alignment, climate policy, and organizational governance, modeling government, industry, and civil society as a policy-space instance of the same recursion. On examination, this claim does not follow from anything proven in this paper or its predecessor: nothing here establishes that real policy processes behave like a coordinate-wise contraction toward a fixed anchor, that real institutions can be modeled as points in $[0,1]^3$, or that convergence in this abstract sense would correspond to good policy outcomes even if the modeling assumption held.

We think the underlying intuition — that three interacting parties, integrating information recursively, might stabilize toward a joint state faster or more predictably under some conditions than others — is not unreasonable as a research question. But turning it into an actual claim would require, at minimum: an explicit, defensible mapping from real institutional behavior to points in a metric space; empirical evidence that such a mapping is accurate; and a normative argument for why the resulting fixed point is desirable, not just reachable. We do not have any of these, and flag AI alignment, climate policy,

and governance here only as directions where the question might be worth posing by researchers with the relevant domain expertise, not as applications this paper delivers.

9. Conclusion

This paper's contribution is Theorem 1: the triadic recursion operator from Series I generalizes to anisotropic per-coordinate blending without losing its unique fixed point, with contraction rate equal to the largest single-coordinate blend parameter, verified computationally. Sections 4 and 6 extend Series I's illustrative applications with a fuller equilibrium characterization and a fully worked engineering example, respectively, both computed rather than asserted. Section 5 remains, honestly, a restatement of Series I's logic analogy with a small addition tied to the new theorem, not a new result. Section 8 explains why we do not believe the framework currently supports any claims about governance or AI alignment, and what would be needed before it could. We think this is a more modest paper than its predecessor draft, and a more accurate one.

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This manuscript was prepared with drafting, mathematical-formalization, and computational-verification assistance from Claude (Anthropic, Claude Sonnet 5). Every numerical and closed-form claim in Sections 3.1, 4, and 6 was computed by executing the operators and matrices exactly as defined in the text; the verification script (`series2_verification.py`) is distributed together with this PDF. The author reviewed all content and is responsible for its claims.

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